

Coursework 2 Report

CIS4514 Programming for Data Science & Artificial Intelligence

Hafeez Ullah Amin

Kate Forshaw

25259628

Word count: 2615

Task 1

For the first task, it was required to examine reviews of video games available on Amazon and discover the ten most popular games, and the ten least popular games. This had to be completed through creating a software on a virtual machine (VM), which should automatically analyse the dataset of Amazon video game reviews.

The Cloudera VM software chosen for this task was Hive (see appendix 1). Hive is a system that outlines, queries, and evaluates large, distinct data sets, and it can be referred to as a data warehouse (Cloudera, 2025). It was implemented as databases can be created within it, emphasising tables may be defined which allows Hive to load data from CSV into the corresponding defined fields (Ronianto et al., 2019). This demonstrates that Hive supports query language (Goodarzi et al), insinuating queries can be designed to investigate the reviews and popularity of Amazon's video games.

A database was created and named video_games (see appendix 2). It then had to be stated to use that database (see appendix 3). A table called reviews was created with three columns: rating as a decimal (2,1) datatype, game_id as varchar (10), and comment also as varchar (1000) (see appendix 4). There was a 'row format delimited fields terminated by' statement within the query to establish that the imported dataset was organised in rows, consisting of multiple pieces of data (IBM, 2025). The dataset of reviews of Amazon's video games was loaded into the table reviews (see appendix 5). To check if all the data was imported, a select query was written to count the total number of reviews, the result was accurate with 40,000 (see appendix 6).

It was necessary to compute the top ten video games with the highest number of 5-star reviews. A query was designed to select game_id and count the number of reviews where rating equalled 5.0, referred to as and ordered by five_star_reviews descending with a limit of 10

games (see appendix 7). This presented the ten games with the highest total of 5-star reviews, for example, the most popular game was B004MC8CA2 with 1605 ratings that were five stars.

Another requirement was to measure the top ten video games with the highest number of 1-star reviews. A query was designed to select `game_id` and count the number of reviews where rating equalled 1.0, referred to as and ordered by `one_star_reviews` descending with a limit of 10 games (see appendix 8). This presented the ten games with the highest total of 1-star reviews, for example, the least popular game was B00409G750 with 78 ratings that were one star.

Some additional queries were ran that may be of interest for Amazon's Research and Development team. They may want to discover the average rating of all games or a specific game. `Game_id` was selected along with the average function being applied to rating, for the game B004LZOTHU, which printed a result of 3.19 rating average (see appendix 9). To view the average ratings of all the video games, remove the 'where' statement, and they will be presented from highest average to lowest via the 'order by' statement.

Another method of accessing rating stats for the video games could be through calculating the distribution of ratings. Rating was selected and counted where `game_id` was B004K5K92K, which demonstrated this game had 9 one-star reviews, 7 two-stars, 10 three-stars, 10 four-stars, and 19 five-stars (see appendix 10). This query was designed for a specific game again, but could relay the distribution of ratings amongst all games by removing the 'where' statement.

Amazon may like to know which games had the greatest number of reviews. `Game_id` was selected and counted for all its reviews, referred to and ordered by as `total_reviews` descending, with a limit of 10 games (see appendix 11). This portrayed a list of the top ten games with the greatest number of reviews, for example, the game B004MC8CA2 had the

highest review total with 2041. If the games with the lowest review totals was of interest, the ‘order by’ statement would need to be converted to ascending, rather than descending.

The Research and Development team may want to view the volume of reviews alongside the average ratings for each game. Game_id was selected and counted – as total_reviews – and average rating was calculated – as average_rating – for the game B004S7XGKW (see appendix 12). This indicated the video game had a total of 5 reviews and an average rating of 2.8 stars. This query could present all video games’ total reviews and average ratings altogether if the ‘where’ statement was removed.

It could be intriguing to discover if there are Amazon video games that have contradicting reviews with a lot of five-star and one-star ratings. Selected was game_id, the sum of ratings equalled to 1 as one_star, and the sum of ratings equalled to 5 as five_star, where both one_star and five_star are more than 50 and ordered by descending, with a limit of 10 (see appendix 13). This proposed a list of the top ten games with the most polarised reviews, for example, the game B004MC8CA2 had 1605 5-star ratings, but also many 1-star ratings (70).

Task 2

After cleaning and reducing the Road Safety Data 2021, it was combined with the metadata to convert meaningless numeric codes to descriptive accident characteristics. This allowed for the visualisation of the accident data to be understood and interpreted more accurately.

A bar chart was plotted to represent number of accidents by collision severity across different road types (see figure 1). It appeared that a single carriageway was more accident prone, especially when the collision was slight. According to the UK Government statistics (2021), A and minor (B, C, and unclassified) roads covered 99% of road length and accounted for 80% of total vehicle miles in 2021. With the vast majority of A, B, and C roads being single carriageways, this implies that accidents were more likely to occur in single carriageways as they were the most common road type and most frequently used. Additionally, slight severity prevailed, followed by serious, then fatal for every road type, which likely would have been the favourable outcome.

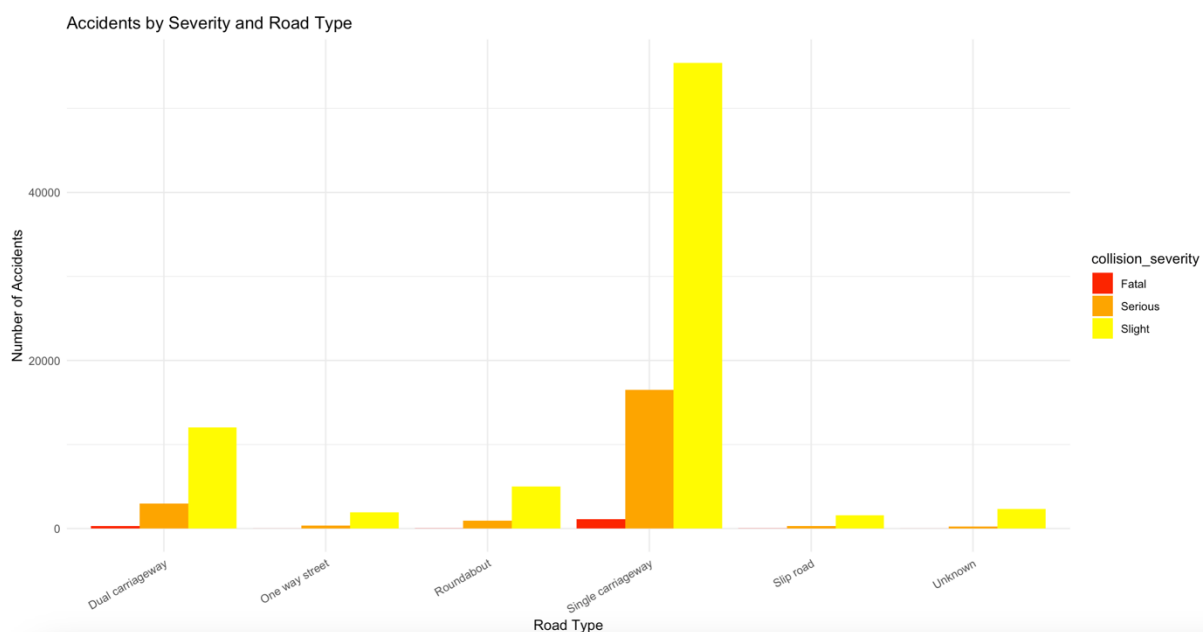


Figure 1 Bar chart representing accidents by severity and road type

Accident Frequency by Weather and Road Surface

Road Surface

Weather Condition

Accident Count

20000
40000
60000

20000
40000
60000

Road Surface	Weather Condition	Accident Count (approx.)
Wet or damp	Fine no high winds	40000
Wet or damp	Raining no high winds	40000
unknown (self reported)	Unknown	20000
Snow	Unknown	20000
Frost or ice	Unknown	20000
Flood over 3cm. deep	Unknown	20000
Dry	Fine no high winds	60000
Dry	Other	20000
Data missing or out of range	Unknown	20000

A dot map demonstrating accident density in Greater Manchester was created (see figure 3). This dot map contains every accident that took place in Greater Manchester in 2021. The approximate coordinates with the most concentration of dots was (-2.25, 53.475), which corresponds to Manchester city centre potentially being the area with the highest accident density. The city centre usually has the highest population in Greater Manchester – in 2021, it

was 551,953, compared to the second highest which was significantly lower at 329,329 (GMCA, 2023). This insinuates that the higher the population of an area in Greater Manchester, the more accidents that are likely to occur.

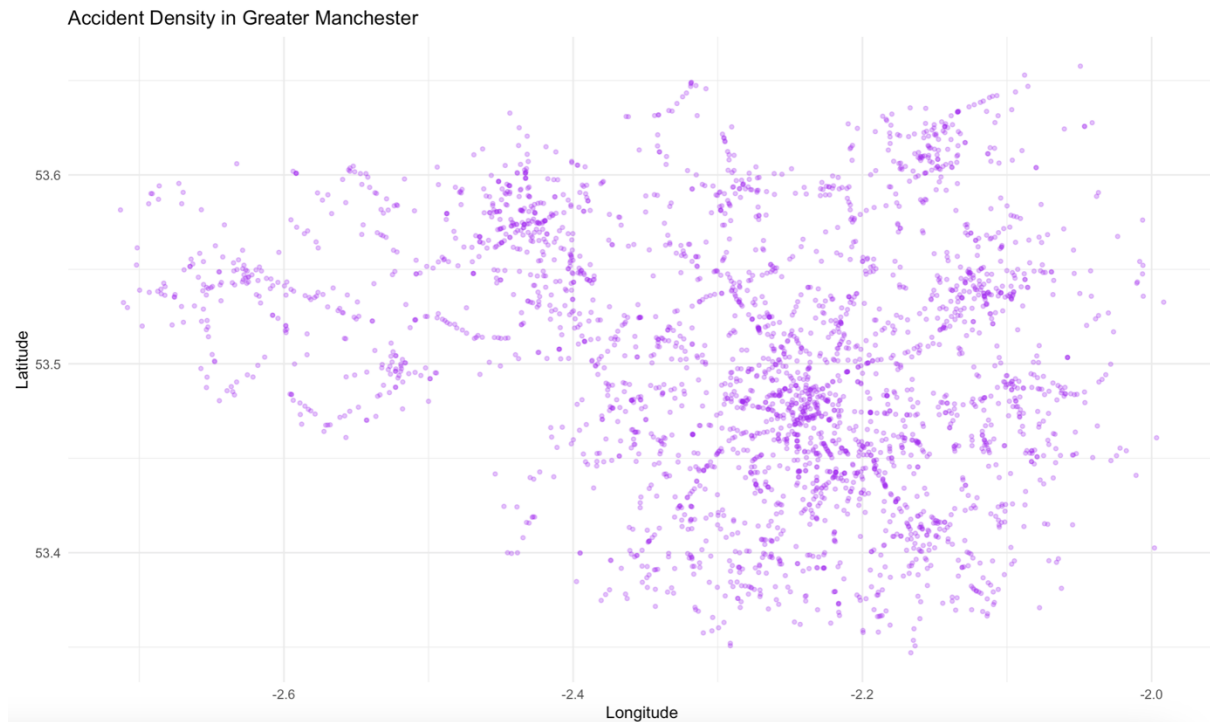


Figure 3 Dot map representing accident density in Greater Manchester

There was a requirement to plot a heatmap or a dot map for Greater Manchester, so a heatmap was represented for accident density in Devon and Cornwall instead (see figure 4). The most concentrated area – portrayed by the darkest point on the heatmap – was the approximate coordinates $(-4.2, 50.38)$, which refers to the town of Plymouth. Plymouth is the largest city in Devon and is right on the boarder of Cornwall, all while being a major waterfront city and urban hub by securing one of the greatest variety of shops (Visit Plymouth, 2025). Due to it being the most urban and central area in Devon and Cornwall, this could be a contributing factor to why the most accidents take place there.

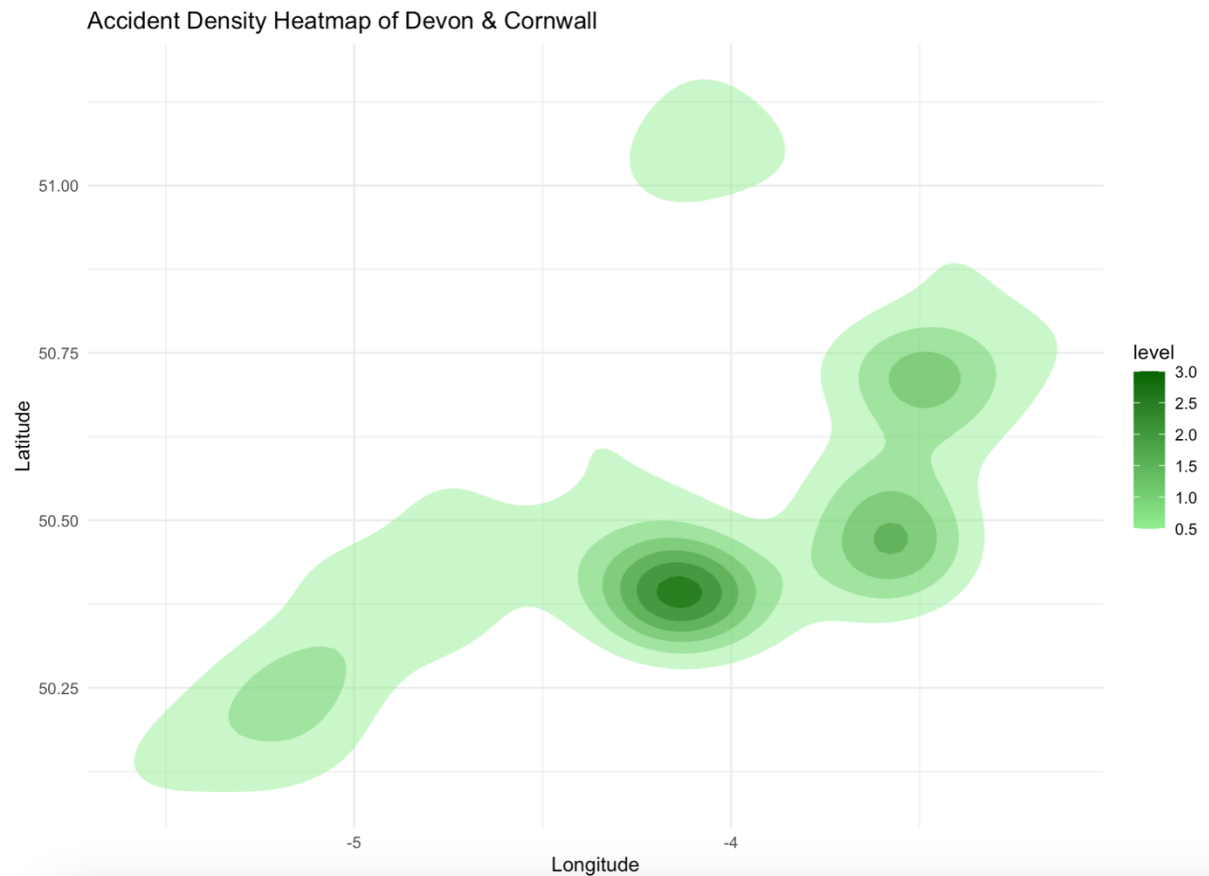


Figure 4 Heat map representing accident density in Devon and Cornwall

It was of interest to discover if there was a relationship between number of casualties and number of vehicles for all the road collision data. Therefore, a line chart was plotted based on the average number of casualties by the number of vehicles involved (see figure 5). This line chart portrayed that the lowest casualty average was 1 with 12 and 13 vehicles being involved – the greatest numbers of vehicles recorded. This was surprising as it infers that there was a potential negative relationship between average casualties and vehicle numbers. However, casualty average was at its highest point at approximately 2.625 when 8 vehicles were involved, and its second lowest point of 1.125 with also the least vehicle involvement of 1. This suggests that a moderate number of vehicles involved in an accident could cause more harm than a significantly small or large number of vehicles.

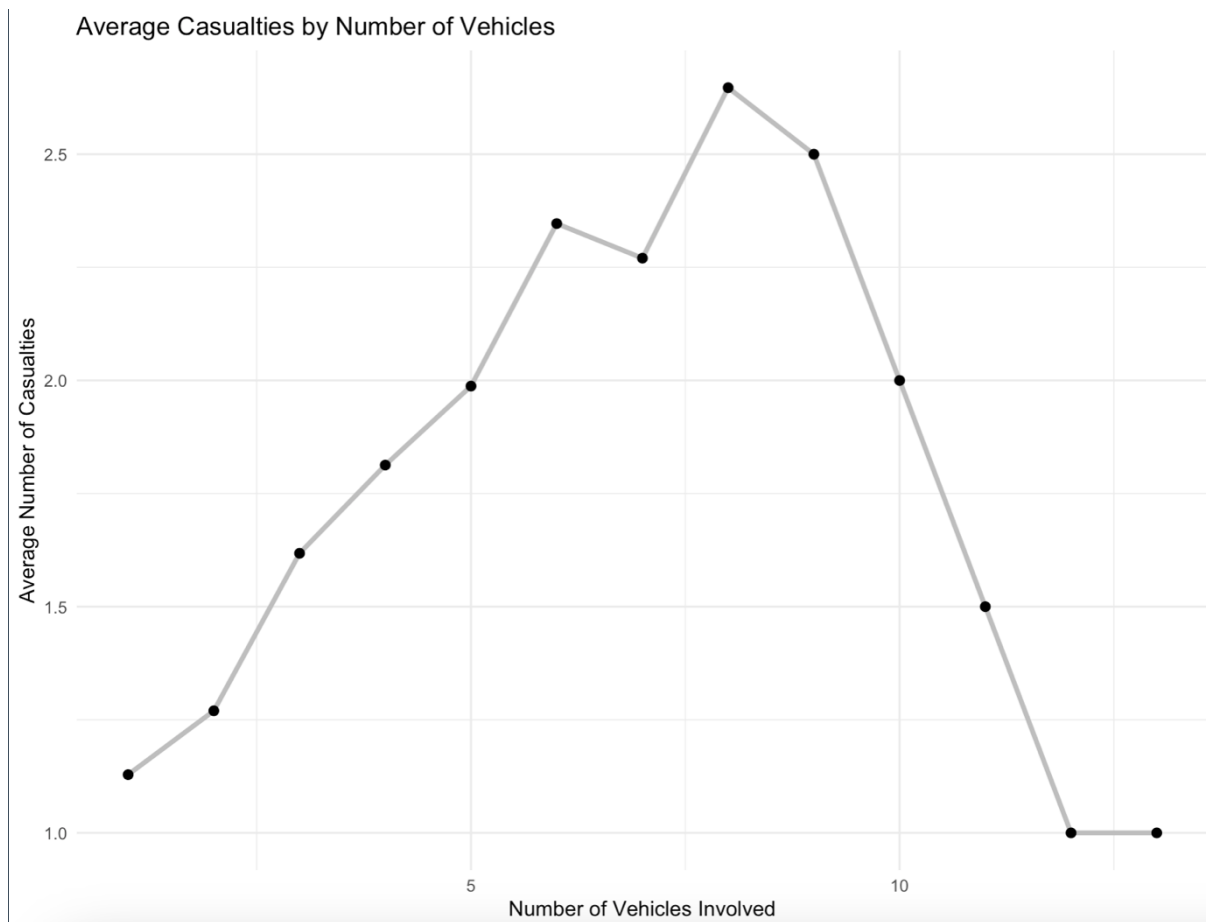


Figure 5 Line chart representing average casualties by number of vehicles

Task 3

For this task, it was required to create an R Shiny interactive dashboard based upon the Road Safety Data 2021. This had to include the accidents dataset alongside either the vehicle or casualty datasets. It was chosen to combine accidents with vehicles, to discover if specific vehicle properties were at fault for the collisions, and if there were any patterns unveiled between vehicles and accidents.

The road accident dataset was loaded into R and inspected using glimpse, summary, and head. Glimpse demonstrated the name, datatype, and first few values of each column, for example, speed limit was a column with integers, which most of the first values were 30 (mph), apart from the third row was 40 (see appendix 14). Summary calculates the minimum, 1st quartile, median, mean, 3rd quartile, and maximum of each integer column. Latitude obtained a mean of 52.36, with the lowest value of 49.98, the highest was 60.5, the coordinate in the centre of all rows was 51.79, 51.46 was halfway in-between the minimum and median, and 53.33 was halfway between the medium and maximum (see appendix 15). Head highlights the first 6 rows of each column, for example, the first six local authority highways presented were E06000001, E06000002, E06000002, E06000003, E06000005, and E06000007 (see appendix 16).

Following this, the road accident dataset was pre-processed through cleaning the names of columns, removing rows with missing values, removing duplicate rows or columns, and removing unnecessary columns. The columns deleted were collision_year, collision_ref_no, location_easting_osgr, location_northing_osgr, local_authority_district, local_authority_highway, first_road_class, first_road_number, second_road_class, second_road_number, pedestrian_crossing_human_control_historic, pedestrian_crossing_physical_facilities_historic, carriageway_hazards_historic,

trunk_road_flag, lsoa_of_accident_location, collision_adjusted_severity_serious, and collision_adjusted_severity_slight. Reasons for these being removed were due to repetitive information, updated information, or simply not required information for trends and insights of interest.

The vehicle dataset was loaded and followed the same process. In glimpse, age of driver was an integer, ten of which were 86, 23, 27, 70, 42, 32, 37, 41, 54, and 4 (see appendix 17). The summary stated age of vehicle had a minimum of -1, 1st quartile of 0, median of 5, mean, of 6.001, 3rd quartile of 11, and a maximum of 91 (see appendix 18). The head presented the first 6 values of engine capacity cc to be 6200, 124, 124, 1968, 1968, and 1798 (see appendix 19). Cleaning the vehicle dataset involved removing collision_year, vehicle_manoeuvre_historic, vehicle_location_restricted_lane_historic, journey_purpose_of_driver_historic, scooter_flag, and driver_distance_banding.

Vehicles were left joined to accidents, so there was one row per vehicle with accident details added. This prevented accidents having duplicated rows. The road safety metadata was obtained, cleaned, and joined to the accident and vehicle combined data to convert meaningless numbers to descriptive labels.

Metrics were created for vehicle types: car, motorcycle, bus, HGV, bike, tractor, tram, mobility scooter, and other. The metrics for vehicle manoeuvre were reversing, parking, waiting, stopping, moving forward, turning, changing lane, overtaking and unknown. Metrics were designed for no skidding, skidded, overturned, jack-knifed, did not hit object and did hit object. Age of driver and vehicle consisted of metrics young driver, middle-aged driver, old driver, old car, and young car. Metrics were also created for propulsion code: petrol, oil, electric, steam, gas, diesel, and other. All the new metrics were inspected through head (see appendix 20). They were produced to help the filtering process for the interactive dashboard.

Before building the interactive dashboard, the column hour had to be specified as numeric as a problem was faced when plotting its values due to its datatype being character. Additionally, a function was designed to calculate the mode of data, which could later be implemented for summary tables.

To build the dashboard, a server had to be implemented. A function was created inside the server to filter the accident vehicle metrics data, so the dashboard would be re-run whenever a filter is altered by a user. The filters chosen were based solely on the metrics created earlier – all based on vehicle characteristics and components. These included vehicle type, manoeuvre, junction location, skidding, hit object, age of driver, age of vehicle, and propulsion.

Tables and graphs were inputted onto the server concerning accident characteristics such as severity, time, weather conditions, and location. The tables were reactive meaning they would change relative to the vehicle filters selected. Whereas, the graphs were reactive and interactive, which allows for insights and representations of points on a graph when hovered over. There was a summary table which provided the total number of crashes, and the mode severity, hour, weather, and local authority district (LAD). The visualisation of the summary table was a lollipop chart representing the mode values, where each mode plotted was assigned a corresponding colour to its individual graphs. There was an individual table created for each of severity, time, weather, and LAD, which contained the count and percent for their distinct variables. A tree map was acquired to plot severity, with red representing fatal, orange for serious, and yellow for slight. The plot for time was a bar chart where each (purple) bar expressed an hour in the day from 0-23. Weather conditions were plotted on a horizontal bar chart in light sky blue. A scatter plot was designed for location, consisting of longitude and latitude coordinates plotted in forest green, to see specifically where in the UK the collision took place.

Following the server set-up was the user interface. This defines what the user views and interacts with. A clean, modern theme called ‘flatly’ was chosen for the dashboard, and it was given a title of ‘UK Collision Trends & Insights 2021’. The sidebar was filled with all the possible filters that could be selected. The main panel had five tabs: overview, severity, time, weather, and location. Overview contained the summary table and lollipop plot; severity contained the severity tree map and breakdown table; time contained the hour bar plot and breakdown table; weather contained the weather bar chart and breakdown table; and location contained the longitude and latitude scatter map and LAD breakdown table. The interactive dashboard was then ready to be ran.

This interactive dashboard should portray patterns between accident characteristics (severity, time, weather, and location) and vehicle characteristics (type, manoeuvre, junction location, skidding, hit object, age of driver, age of vehicle, and propulsion) involved in a crash. It was a requirement to represent location by regions in the UK, however, there was difficulty faced. There were several attempts to join region name datasets from different years with the accidents and vehicle dataset, but it was not successful as the names of LADs did not accurately align between the two datasets. Therefore, location trends were sufficiently covered by both LAD and coordinates.

References

CLOUDERA, 2025. Apache Hive key features. *Cloudera.com* [online]. Available from:

https://docs-archive.cloudera.com/runtime/7.0.3/hive-introduction/topics/hive_whats_new_in_this_release_hive.html

GOV UK, 2021. Road Safety Data – Accidents 2021. *Gov.uk* [online]. Available from:

<https://data.dft.gov.uk/road-accidents-safety-data/dft-road-casualty-statistics-collision-2021.csv>

GOV UK, 2021. Road Safety Data – Vehicles 2021. *Gov.uk* [online]. Available from:

<https://share.google/mZesT4vOR5bWVN6nN>

GOV UK, 2024. Road Safety Data Guide. *Gov.uk* [online]. Available from:

<https://data.dft.gov.uk/road-accidents-safety-data/dft-road-casualty-statistics-road-safety-open-dataset-data-guide-2024.xlsx>

GOV UK, 2024. Road traffic estimates in Great Britain, 2024: Traffic in Great Britain by road type. *Gov.uk* [online]. Available from:

<https://www.gov.uk/government/statistics/road-traffic-estimates-in-great-britain-2024/road-traffic-estimates-in-great-britain-2024-traffic-in-great-britain-by-road-type>

GOV UK, 2025. Guidance on road classification and the primary route

network. *Gov.uk* [online]. Available from:

<https://www.gov.uk/government/publications/guidance-on-road-classification-and-the-primary-route-network/guidance-on-road-classification-and-the-primary-route-network>

GREATER MANCHESTER COMBINED AUTHORITY (GMCA), 2023. Census 2021

Briefing – Total population. *Gov.uk* [online]. Available from:

https://www.greatermanchester-ca.gov.uk/media/7869/230514_population_final.pdf

IBM, 2025. IBM TRIRIGA application platform. *Ibm.com* [online]. Available from:

<https://www.ibm.com/docs/en/tap/4.4.0?topic=process-tab-delimited-files>

PINSKWAR, I., CHORYNSKI, A. and GRACZYK, D., 2024. Good weather for a ride (or not?): how weather conditions impact road accidents—a case study from Wielkopolska (Poland). *International journal of biometeorology*, 68(2), pp.317-331.

Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10794278/>

RONIANTO, M.F., ASROR, I., PRABOWO, S. and NUGRAHA, F.A., 2019, March.

Performance evaluation sql-on-hadoop: a case study of Hortonworks and Cloudera.

In *Journal of Physics: Conference Series* (Vol. 1192, No. 1, p. 012016). IOP

Publishing. Available from: <https://doi.org/10.1088/1742-6596/1192/1/012016>

VISIT PLYMOUTH, 2025. Plymouth – Britain’s Ocean City. *Visit Plymouth* [online].

Available from: <https://www.visitplymouth.co.uk/explore/areas-to-visit/devon>

Appendices

Appendix 1 Implement Hive software on Cloudera

```
[cloudera@quickstart ~]$ hive
```

```
Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.p
roperties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
```

Appendix 2 Create video games database

```
hive> create database video_games;
OK
Time taken: 0.464 seconds
```

Appendix 3 Use video games database

```
hive> use video_games;
OK
Time taken: 0.064 seconds
```

Appendix 4 Create reviews table

```
hive> create table reviews (rating decimal(2,1), game_id varchar(10), comment va
rchar(1000))
> row format delimited fields terminated by '\t';
OK
Time taken: 1.639 seconds
```

Appendix 5 Load Amazon video game reviews dataset

```
hive> load data inpath 'CIS4514 CW 2.1 Data.csv' into table reviews;
Loading data to table video_games.reviews
Table video_games.reviews stats: [numFiles=1, totalSize=10415366]
OK
Time taken: 1.267 seconds
```

Appendix 6 Count total number of reviews

```
hive> select count(*) from reviews;
Query ID = cloudera_20251216050707_2be01d20-17b4-49bb-ae71-516e8f44e163
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0018, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0018/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0018
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-16 05:08:13,947 Stage-1 map = 0%, reduce = 0%
2025-12-16 05:08:23,547 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 3.65 sec
2025-12-16 05:08:30,941 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 5.45 sec
MapReduce Total cumulative CPU time: 5 seconds 450 msec
Ended Job = job_1765882425567_0018
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 5.45 sec HDFS Read: 1042352
5 HDFS Write: 6 SUCCESS
Total MapReduce CPU Time Spent: 5 seconds 450 msec
OK
40000
Time taken: 34.995 seconds, Fetched: 1 row(s)
```

Appendix 7 Select games and count five-star reviews

```
hive> select game_id, count(*) as five_star_reviews
> from reviews
> where rating = 5.0
> group by game_id
> order by five_star_reviews
> desc
> limit 10;
Query ID = cloudera_20251216051212_119112dd-8869-4a4b-b581-35fba4188696
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0019, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0019/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0019
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-16 05:12:19,269 Stage-1 map = 0%, reduce = 0%
2025-12-16 05:12:28,779 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 8.38 sec
2025-12-16 05:12:35,134 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 10.23 sec
MapReduce Total cumulative CPU time: 10 seconds 230 msec
Ended Job = job_1765882425567_0019
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0020, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0020/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0020

Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-12-16 05:12:44,821 Stage-2 map = 0%, reduce = 0%
2025-12-16 05:12:51,135 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 1.4 sec
2025-12-16 05:12:58,460 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 3.2 sec
MapReduce Total cumulative CPU time: 3 seconds 200 msec
Ended Job = job_1765882425567_0020
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 10.23 sec HDFS Read: 104238
37 HDFS Write: 15802 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 3.2 sec HDFS Read: 20992 HD
FS Write: 151 SUCCESS
Total MapReduce CPU Time Spent: 13 seconds 430 msec
OK
B004MC8CA2      1605
B004HGK6FW      723
B004H6WTJI      655
B004SBS8LA      609
B004HXHVZ8      495
B004LOMB2Q      484
B004Q3CJQ0      436
B004HE5TAG      427
B004K4RY9M      394
B004DLPXA0      387
Time taken: 50.416 seconds, Fetched: 10 row(s)
```

Appendix 8 Select games and count one-star reviews

```
hive> select game_id, count(*) as one_star_reviews
> from reviews
> where rating = 1.0
> group by game_id
> order by one_star_reviews
> desc
> limit 10;
Query ID = cloudera_20251216051717_6aeaaa5c-2504-456c-9f27-ee61a05b2347
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0021, Tracking URL = http://quickstart.cloudera
:8088/proxy/application_1765882425567_0021/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0021
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-16 05:17:40,964 Stage-1 map = 0%, reduce = 0%
2025-12-16 05:17:50,499 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 7.28 se
c
2025-12-16 05:17:56,804 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 9.11
sec
MapReduce Total cumulative CPU time: 9 seconds 110 msec
Ended Job = job_1765882425567_0021
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0022, Tracking URL = http://quickstart.cloudera
:8088/proxy/application_1765882425567_0022/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0022

Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-12-16 05:18:07,259 Stage-2 map = 0%, reduce = 0%
2025-12-16 05:18:14,794 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 1.49 se
c
2025-12-16 05:18:22,170 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 3.5 s
ec
MapReduce Total cumulative CPU time: 3 seconds 500 msec
Ended Job = job_1765882425567_0022
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 9.11 sec HDFS Read: 1042383
6 HDFS Write: 13846 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 3.5 sec HDFS Read: 19025 HD
FS Write: 140 SUCCESS
Total MapReduce CPU Time Spent: 12 seconds 610 msec
OK
B00409G750      78
B004DPCSKI      75
B004FOA84A      71
B004MC8CA2      70
B004SBS8LA      66
B004QJKRZ4      65
B004DPIEF6      62
B004LZIZHU      61
B004HE5TAG      59
B004RJMUJO      56
Time taken: 52.178 seconds, Fetched: 10 row(s)
```

Appendix 9 Select games and average ratings

```
hive> select game_id, avg(rating) as average_rating
> from reviews
> where game_id = 'B004LZ0THU'
> group by game_id
> order by average_rating desc;
Query ID = cloudera_20251216052626_179b0f15-276b-4491-9a77-302565c96db9
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0025, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0025/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0025
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-16 05:26:57,383 Stage-1 map = 0%, reduce = 0%
2025-12-16 05:27:07,895 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 4.46 sec
2025-12-16 05:27:14,212 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 6.38 sec
MapReduce Total cumulative CPU time: 6 seconds 380 msec
Ended Job = job_1765882425567_0025
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0026, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0026/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0026
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-12-16 05:27:23,455 Stage-2 map = 0%, reduce = 0%

2025-12-16 05:27:30,954 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 1.22 sec
2025-12-16 05:27:38,299 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 3.02 sec
MapReduce Total cumulative CPU time: 3 seconds 20 msec
Ended Job = job_1765882425567_0026
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 6.38 sec HDFS Read: 1042371
5 HDFS Write: 129 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 3.02 sec HDFS Read: 5283 HD
FS Write: 19 SUCCESS
Total MapReduce CPU Time Spent: 9 seconds 400 msec
OK
B004LZ0THU 3.18519
Time taken: 50.651 seconds, Fetched: 1 row(s)
```

Appendix 10 Select and count ratings

```
hive> select rating, count(*) as count
> from reviews
> where game_id = 'B004K5K92K'
> group by rating
> order by rating;
Query ID = cloudera_20251216053131_47e09df6-7478-4faf-8660-40b139f5481e
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0027, Tracking URL = http://quickstart.cloudera
:8088/proxy/application_1765882425567_0027/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0027
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-16 05:32:05,914 Stage-1 map = 0%, reduce = 0%
2025-12-16 05:32:15,542 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 4.75 se
c
2025-12-16 05:32:22,915 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 6.98
sec
MapReduce Total cumulative CPU time: 6 seconds 980 msec
Ended Job = job_1765882425567_0027
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0028, Tracking URL = http://quickstart.cloudera
:8088/proxy/application_1765882425567_0028/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0028
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-12-16 05:32:32,231 Stage-2 map = 0%, reduce = 0%

2025-12-16 05:32:38,536 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 1.27 se
c
2025-12-16 05:32:45,874 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 3.08
sec
MapReduce Total cumulative CPU time: 3 seconds 80 msec
Ended Job = job_1765882425567_0028
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 6.98 sec HDFS Read: 1042382
7 HDFS Write: 201 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 3.08 sec HDFS Read: 5247 HD
FS Write: 23 SUCCESS
Total MapReduce CPU Time Spent: 10 seconds 60 msec
OK
1      9
2      7
3     10
4     10
5     19
Time taken: 50.956 seconds, Fetched: 5 row(s)
```

Appendix 11 Select games and count total reviews

```
hive> select game_id, count(*) as total_reviews
> from reviews
> group by game_id
> order by total_reviews desc
> limit 10;
Query ID = cloudera_20251216053535_f4df759f-8ed7-49ad-b6bc-486b4a720466
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0029, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0029/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0029
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-16 05:35:53,129 Stage-1 map = 0%, reduce = 0%
2025-12-16 05:36:01,533 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 7.3 sec
2025-12-16 05:36:07,891 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 9.27 sec
MapReduce Total cumulative CPU time: 9 seconds 270 msec
Ended Job = job_1765882425567_0029
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0030, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0030/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0030
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-12-16 05:36:19,389 Stage-2 map = 0%, reduce = 0%
2025-12-16 05:36:26,961 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 1.65 se
c
2025-12-16 05:36:34,336 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 3.58 sec
MapReduce Total cumulative CPU time: 3 seconds 580 msec
Ended Job = job_1765882425567_0030
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 9.27 sec HDFS Read: 1042302
9 HDFS Write: 16414 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 3.58 sec HDFS Read: 21595 H
DFS Write: 152 SUCCESS
Total MapReduce CPU Time Spent: 12 seconds 850 msec
OK
B004MC8CA2      2041
B004HGK6FW      1066
B004SBS8LA      959
B004K4RY9M      906
B004HE5TAG      905
B004H6WTJI      899
B004HXHVZ8      720
B004LOMB2Q      719
B00409G750      688
B004LZIZHU      610
Time taken: 51.057 seconds, Fetched: 10 row(s)
```

Appendix 12 Select games and average ratings and, count total reviews

```
hive> select game_id, count(*) as total_reviews, avg(rating) as average_rating
> from reviews
> where game_id = 'B004S7XGKW'
> group by game_id
> order by total_reviews desc, average_rating desc;
Query ID = cloudera_20251216054545_5775917c-0981-4862-96c9-c5b078246a70
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0031, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0031/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0031
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-16 05:45:30,183 Stage-1 map = 0%, reduce = 0%
2025-12-16 05:45:38,580 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 4.16 sec
2025-12-16 05:45:45,927 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 6.03 sec
MapReduce Total cumulative CPU time: 6 seconds 30 msec
Ended Job = job_1765882425567_0031
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0032, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1765882425567_0032/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0032
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-12-16 05:45:55,689 Stage-2 map = 0%, reduce = 0%
```

Appendix 13 Select games and sum one-star and five-star reviews

```
hive> select game_id, sum(case when rating = 1 then 1 else 0 end) as one_star, s
um(case when rating = 5 then 1 else 0 end) as five_star
  > from reviews
  > group by game_id
  > having one_star > 50 and five_star > 50
  > order by five_star desc, one_star desc
  > limit 10;
Query ID = cloudera_20251216055252_198cc219-1db7-49bd-a4e7-9c582b4553d1
Total jobs = 2
Launching Job 1 out of 2
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0033, Tracking URL = http://quickstart.cloudera
:8088/proxy/application_1765882425567_0033/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0033
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-12-16 05:52:48,585 Stage-1 map = 0%, reduce = 0%
2025-12-16 05:52:57,026 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 5.08 se
c
2025-12-16 05:53:05,433 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 7.86
sec
MapReduce Total cumulative CPU time: 7 seconds 860 msec
Ended Job = job_1765882425567_0033
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1765882425567_0034, Tracking URL = http://quickstart.cloudera
:8088/proxy/application_1765882425567_0034/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1765882425567_0034

Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-12-16 05:53:15,985 Stage-2 map = 0%, reduce = 0%
2025-12-16 05:53:23,353 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 1.46 se
c
2025-12-16 05:53:29,825 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 3.34
sec
MapReduce Total cumulative CPU time: 3 seconds 340 msec
Ended Job = job_1765882425567_0034
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 7.86 sec HDFS Read: 1042481
2 HDFS Write: 442 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 3.34 sec HDFS Read: 6247 HD
FS Write: 181 SUCCESS
Total MapReduce CPU Time Spent: 11 seconds 200 msec
OK
B004MC8CA2      70      1605
B004SBS8LA      66      609
B004HE5TAG      59      427
B004SBQGHS      53      337
B00409G750      78      278
B004LZIZHU      61      236
B004DPCSKI      75      225
B004RJMUIJ0     56      220
B004FOA84A      71      189
B004DPIEF6      62      168
Time taken: 50.19 seconds, Fetched: 10 row(s)
```

Appendix 14 Glimpse of road accident dataset

```
> glimpse(road_accident_dataset)
Rows: 101,087
Columns: 44
$ collision_index          <chr> "2021170H10581", "2021170M1018...
$ collision_year           <int> 2021, 2021, 2021, 2021, 2021, ...
$ collision_ref_no         <chr> "170H10581", "170M10181", "170...
$ location_easting_osgr    <int> 450648, 448821, 451070, 471589...
$ location_northing_osgr  <int> 534766, 519183, 514313, 518355...
$ longitude               <dbl> -1.215535, -1.246477, -1.21254...
$ latitude                <dbl> 54.70538, 54.56554, 54.52156, ...
$ police_force            <int> 17, 17, 17, 17, 11, 7, 4, 16, ...
$ collision_severity       <int> 3, 2, 3, 2, 3, 2, 2, 3, 3, 2, ...
$ number_of_vehicles       <int> 2, 1, 2, 1, 2, 2, 1, 1, 2, 2, ...
$ number_of_casualties    <int> 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, ...
$ date                    <chr> "07/07/2021", "26/02/2021", "2...
$ day_of_week             <int> 4, 6, 5, 7, 3, 4, 3, 6, 2, 1, ...
$ time                    <chr> "12:30", "19:30", "16:45", "18...
$ local_authority_district <int> -1, 243, -1, 241, -1, -1, -1, ...
$ local_authority_ons_district <chr> "E06000001", "E06000002", "E06...
$ local_authority_highway  <chr> "E06000001", "E06000002", "E06...
$ local_authority_highway_current <chr> "E06000001", "E06000002", "E06...
$ first_road_class         <int> 6, 6, 5, 3, 6, 3, 6, 6, 3, 3, ...
$ first_road_number        <int> 0, 0, 0, 174, 0, 57, 0, 0, 107...
$ road_type               <int> 6, 6, 1, 6, 6, 3, 6, 6, 3, 3, ...
$ speed_limit              <int> 30, 30, 40, 30, 30, 30, 30, 30...
$ junction_detail_historic <int> 3, 3, 1, 3, 3, 3, 0, 3, 6, 3, ...
$ junction_detail          <int> 13, 13, 0, 13, 13, 13, 0, 13, ...
$ junction_control         <int> 4, 4, 4, 4, 4, 4, -1, 4, 2, 4,...
$ second_road_class        <int> 6, 6, 6, 6, 6, 6, 0, 6, 6, 6, ...
$ second_road_number       <int> 0, 0, 0, 0, 0, 0, -1, 0, 0, 0,...
$ pedestrian_crossing_human_control_historic <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
$ pedestrian_crossing_physical_facilities_historic <int> 0, 4, 0, 0, 0, 0, 8, 0, 4, 0, ...
$ pedestrian_crossing      <int> 0, 14, 0, 0, 0, 0, 17, 0, 14, ...
$ light_conditions         <int> 1, 4, 1, 4, 4, 1, 4, 4, 1, 1, ...
$ weather_conditions       <int> 1, 1, 2, 1, 2, 1, 5, 1, 1, 1, ...
$ road_surface_conditions  <int> 1, 1, 2, 1, 2, 1, 2, 4, 1, 1, ...
$ special_conditions_at_site <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
```

```
$ carriageway_hazards_historic <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
$ carriageway_hazards        <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
$ urban_or_rural_area        <int> 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, ...
$ did_police_officer_attend_scene_of_accident <int> 2, 2, 1, 1, 1, 1, 1, 2, 2, 1, ...
$ trunk_road_flag            <int> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, ...
$ lsoa_of_accident_location  <chr> "E01011992", "E01012040", "E01...
$ enhanced_severity_collision <int> -1, -1, -1, -1, 3, -1, 6, 3, 3...
$ collision_injury_based     <int> 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, ...
$ collision_adjusted_severity_serious <dbl> 0.10505612, 1.00000000, 0.0285...
$ collision_adjusted_severity_slight <dbl> 0.8949439, 0.0000000, 0.971487...
```

Appendix 15 Summary of road accident dataset

```
> summary(road_accident_dataset)
collision_index      collision_year collision_ref_no  location_easting_osgr
Length:101087      Min.   :2021      Length:101087      Min.   : 67564
Class :character    1st Qu.:2021      Class :character    1st Qu.:392252
Mode  :character    Median :2021      Mode  :character    Median :461832
                        Mean   :2021                        Mean   :455371
                        3rd Qu.:2021                        3rd Qu.:530038
                        Max.   :2021                        Max.   :655140
                        NA's   :17                        NA's   :17

location_northing_osgr longitude      latitude      police_force
Min.   : 13898      Min.   :-7.4869      Min.   :49.98      Min.   : 1.00
1st Qu.: 175175      1st Qu.: -2.1167      1st Qu.:51.46      1st Qu.: 4.00
Median : 211642      Median : -1.0872      Median :51.79      Median :21.00
Mean   : 274267      Mean   : -1.2050      Mean   :52.36      Mean   :27.06
3rd Qu.: 382390      3rd Qu.: -0.1275      3rd Qu.:53.33      3rd Qu.:44.00
Max.   :1179892      Max.   : 1.7560      Max.   :60.50      Max.   :99.00
NA's   :17          NA's   :17          NA's   :17

collision_severity number_of_vehicles number_of_casualties      date
Min.   :1.00      Min.   : 1.000      Min.   : 1.000      Length:101087
1st Qu.:3.00      1st Qu.: 1.000      1st Qu.: 1.000      Class :character
Median :3.00      Median : 2.000      Median : 1.000      Mode  :character
Mean   :2.76      Mean   : 1.844      Mean   : 1.268
3rd Qu.:3.00      3rd Qu.: 2.000      3rd Qu.: 1.000
Max.   :3.00      Max.   :13.000      Max.   :22.000

      day_of_week      time      local_authority_district
Min.   :1.000      Length:101087      Min.   : -1.0000
1st Qu.:2.000      Class :character    1st Qu.: -1.0000
Median :4.000      Mode  :character    Median : -1.0000
Mean   :4.139                        Mean   : -0.3518
3rd Qu.:6.000                        3rd Qu.: -1.0000
Max.   :7.000                        Max.   :480.0000

local_authority_ons_district local_authority_highway local_authority_highway_current
Length:101087      Length:101087      Length:101087
Class :character    Class :character    Class :character
Mode  :character    Mode  :character    Mode  :character

first_road_class first_road_number      road_type      speed_limit
Min.   :1.000      Min.   : 0.0      Min.   :1.000      Min.   :20
1st Qu.:3.000      1st Qu.: 0.0      1st Qu.:6.000      1st Qu.:30
Median :4.000      Median : 34.0      Median :6.000      Median :30
Mean   :4.205      Mean   : 789.6      Mean   :5.257      Mean   :36
3rd Qu.:6.000      3rd Qu.: 532.0      3rd Qu.:6.000      3rd Qu.:40
Max.   :6.000      Max.   :9480.0      Max.   :9.000      Max.   :70

junction_detail_historic junction_detail junction_control second_road_class
Min.   : -1.000      Min.   : -1.000      Min.   : -1.000      Min.   : -1.000
1st Qu.: 0.000      1st Qu.: 0.000      1st Qu.: -1.000      1st Qu.: 0.000
Median : 2.000      Median : 0.000      Median : 2.000      Median : 3.000
Mean   : 4.386      Mean   : 6.811      Mean   : 1.771      Mean   : 3.088
3rd Qu.: 3.000      3rd Qu.:13.000      3rd Qu.: 4.000      3rd Qu.: 6.000
Max.   :99.000      Max.   :19.000      Max.   : 9.000      Max.   : 6.000

second_road_number pedestrian_crossing_human_control_historic
Min.   : -1.0      Min.   : -1.0000
1st Qu.: -1.0      1st Qu.: 0.0000
Median : 0.0      Median : 0.0000
Mean   : 223.7      Mean   : 0.3621
3rd Qu.: 0.0      3rd Qu.: 0.0000
Max.   :9176.0      Max.   : 9.0000

pedestrian_crossing_physical_facilities_historic pedestrian_crossing light_conditions
Min.   : -1.000                        Min.   : -1.000      Min.   : -1.000
1st Qu.: 0.000                        1st Qu.: 0.000      1st Qu.: 1.000
Median : 0.000                        Median : 0.000      Median : 1.000
Mean   : 1.168                        Mean   : 5.392      Mean   : 1.977
3rd Qu.: 0.000                        3rd Qu.: 0.000      3rd Qu.: 4.000
Max.   : 9.000                        Max.   :99.000      Max.   : 7.000

weather_conditions road_surface_conditions special_conditions_at_site
Min.   : -1.000      Min.   : -1.000      Min.   : -1.0000
1st Qu.: 1.000      1st Qu.: 1.000      1st Qu.: 0.0000
Median : 1.000      Median : 1.000      Median : 0.0000
Mean   : 1.651      Mean   : 1.347      Mean   : 0.2531
```

```

3rd Qu.: 1.000    3rd Qu.: 2.000    3rd Qu.: 0.0000
Max.      : 9.000    Max.      : 9.000    Max.      : 9.0000

carriageway_hazards_historic carriageway_hazards urban_or_rural_area
Min.      :-1.0000    Min.      :-1.000    Min.      :1.00
1st Qu.   : 0.0000    1st Qu.   : 0.000    1st Qu.   :1.00
Median    : 0.0000    Median    : 0.000    Median    :1.00
Mean      : 0.1966    Mean      : 1.791    Mean      :1.32
3rd Qu.   : 0.0000    3rd Qu.   : 0.000    3rd Qu.   :2.00
Max.      : 9.0000    Max.      :99.000    Max.      :3.00

did_police_officer_attend_scene_of_accident trunk_road_flag lsoa_of_accident_location
Min.      :1.000    Min.      :-1.000    Length:101087
1st Qu.   :1.000    1st Qu.   : 2.000    Class :character
Median    :1.000    Median    : 2.000    Mode  :character
Mean      :1.451    Mean      : 1.722
3rd Qu.   :2.000    3rd Qu.   : 2.000
Max.      :3.000    Max.      : 2.000

enhanced_severity_collision collision_injury_based collision_adjusted_severity_severity
Min.      :-1.000    Min.      :0.0000    Min.      :0.0000
1st Qu.   :-1.000    1st Qu.   :0.0000    1st Qu.   :0.0000
Median    :-1.000    Median    :0.0000    Median    :0.0000
Mean      : 1.392    Mean      :0.4962    Mean      :0.2295
3rd Qu.   : 3.000    3rd Qu.   :1.0000    3rd Qu.   :0.1422
Max.      : 7.000    Max.      :1.0000    Max.      :1.0000

collision_adjusted_severity_slight
Min.      :0.0000
1st Qu.   :0.8346
Median    :1.0000
Mean      :0.7560
3rd Qu.   :1.0000
Max.      :1.0000

```

Appendix 16 Head of road accident dataset

```

> head(road_accident_dataset)
  collision_index collision_year collision_ref_no location_easting_osgr
1  2021170H10581         2021      170H10581      450648
2  2021170M10181         2021      170M10181      448821
3  2021170M30721         2021      170M30721      451070
4  2021170L50121         2021      170L50121      471589
5  2021111130730         2021      111130730      429225
6  2021070542272         2021      070542272      362471

  location_northing_osgr longitude latitude police_force collision_severity
1          534766 -1.215535 54.70538         17          3
2          519183 -1.246477 54.56554         17          2
3          514313 -1.212540 54.52156         17          3
4          518355 -0.894618 54.55540         17          2
5          515059 -1.549932 54.52998         11          3
6          389091 -2.565859 53.39732          7          2

  number_of_vehicles number_of_casualties date day_of_week time
1          2          1  1 07/07/2021          4 12:30
2          1          1  1 26/02/2021          6 19:30
3          2          1  1 20/05/2021          5 16:45
4          1          1  1 30/01/2021          7 18:08
5          2          2  2 19/10/2021          3 19:44
6          2          1  1 04/08/2021          4 13:08

  local_authority_district local_authority_ons_district local_authority_highway
1          -1          E06000001          E06000001
2          243          E06000002          E06000002
3          -1          E06000002          E06000002
4          241          E06000003          E06000003
5          -1          E06000005          E06000005
6          -1          E06000007          E06000007

  local_authority_highway_current first_road_class first_road_number road_type
1          E06000001          6          0          6
2          E06000002          6          0          6
3          E06000002          5          0          1
4          E06000003          3          174          6
5          E06000005          6          0          6
6          E06000007          3          57          3

  speed_limit junction_detail_historic junction_detail junction_control

```

1	30	3	13	4
2	30	3	13	4
3	40	1	0	4
4	30	3	13	4
5	30	3	13	4
6	30	3	13	4
second_road_class second_road_number pedestrian_crossing_human_control_historic				
1	6	0		0
2	6	0		0
3	6	0		0
4	6	0		0
5	6	0		0
6	6	0		0
pedestrian_crossing_physical_facilities_historic pedestrian_crossing light_conditions				
1			0	0
2			4	14
3			0	0
4			0	0
5			0	0
6			0	0
weather_conditions road_surface_conditions special_conditions_at_site				
1	1	1		0
2	1	1		0
3	2	2		0
4	1	1		0
5	2	2		0
6	1	1		0
carriageway_hazards_historic carriageway_hazards urban_or_rural_area				
1		0	0	1
2		0	0	1
3		0	0	1
4		0	0	2
5		0	0	1
6		0	0	1
did_police_officer_attend_scene_of_accident trunk_road_flag lsoa_of_accident_location				
1		2	2	E01011992
2		2	2	E01012040

3		1	2	E01012033
4		1	2	E01012136
5		1	2	E01033483
6		1	2	E01012478
enhanced_severity_collision collision_injury_based collision_adjusted_severity_serious				
1	-1		0	0.10505612
2	-1		0	1.00000000
3	-1		0	0.02851205
4	-1		0	1.00000000
5	3		1	0.00000000
6	-1		0	1.00000000
collision_adjusted_severity_slight				
1				0.8949439
2				0.0000000
3				0.9714879
4				0.0000000
5				1.0000000
6				0.0000000

Appendix 17 Glimpse of vehicle dataset

```
> glimpse(vehicle_dataset)
Rows: 186,443
Columns: 32
$ collision_index      <chr> "2021371073032", "2021010330300", "20...
$ collision_year       <int> 2021, 2021, 2021, 2021, 2021, 2021, 2...
$ collision_ref_no     <chr> "371073032", "010330300", "010331714"...
$ vehicle_reference    <int> 1, 1, 1, 2, 2, 2, 4, 2, 2, 2, 2, 1, 1...
$ vehicle_type         <int> 9, 3, 3, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9...
$ towing_and_articulation <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -1, ...
$ vehicle_manoeuvre_historic <int> 18, 9, 99, 17, 4, 2, 18, 2, 18, 5, 18...
$ vehicle_manoeuvre    <int> 19, 9, 99, 19, 4, 2, 19, 2, 19, 5, 19...
$ vehicle_direction_from <int> 7, 3, 9, 7, 7, 0, 1, 0, 1, 5, 6, 7, 3...
$ vehicle_direction_to <int> 3, 1, 9, 4, 3, 0, 5, 0, 5, 1, 2, 3, 1...
$ vehicle_location_restricted_lane_historic <int> 0, 0, 99, 0, 0, 0, 0, 9, 0, 0, 0, 0, ...
$ vehicle_location_restricted_lane <int> 0, 0, 99, 0, 0, 0, 0, 9, 0, 0, 0, 0, ...
$ junction_location    <int> 0, 8, 9, 1, 0, 0, 0, 1, 1, 8, 1, 1, 2...
$ skidding_and_overturning <int> 1, 5, 9, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ hit_object_in_carriageway <int> 0, 0, 99, 0, 0, 0, 0, 4, 0, 0, 0, 0, ...
$ vehicle_leaving_carriageway <int> 1, 0, 9, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ hit_object_off_carriageway <int> 0, 0, 99, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
$ first_point_of_impact <int> 4, 4, 1, 1, 2, 2, 1, 1, 1, 3, 1, 1, 1...
$ vehicle_left_hand_drive <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ journey_purpose_of_driver_historic <int> 5, 2, 2, 5, 5, 5, 2, 6, 6, 6, 6, 3, 2...
$ journey_purpose_of_driver <int> 6, 2, 2, 6, 6, 6, 2, 6, 6, 6, 6, 7, 2...
$ sex_of_driver         <int> 1, 1, 1, 1, 2, 2, 1, 1, 2, 2, 3, 2, 2...
$ age_of_driver         <int> 86, 23, 27, 70, 42, 32, 37, 41, 54, 4...
$ age_band_of_driver    <int> 11, 5, 6, 10, 7, 6, 7, 7, 8, 8, -1, 8...
$ engine_capacity_cc     <int> 6200, 124, 124, 1968, 1998, 1798, 199...
$ propulsion_code        <int> 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 2, 1, 1...
$ age_of_vehicle         <int> 48, 6, 7, 14, 13, 13, 13, 14, 15, 14,...
$ generic_make_model     <chr> "-1", "-1", "-1", "DODGE CALIBER", "D...
$ driver_imd_decile      <int> 9, 9, 3, 3, 2, 8, -1, 2, 1, 2, -1, 1,...
$ lsoa_of_driver         <chr> "E01021681", "E01004079", "E01000535"...
$ scooter_flag           <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ driver_distance_banding <int> -1, -1, -1, -1, -1, -1, -1, -1, -1, -...
```

```
> summary(vehicle_dataset)
```

collision_index	collision_year	collision_ref_no	vehicle_reference	vehicle_type
Length:186443	Min. :2021	Length:186443	Min. : 1.000	Min. :-1.000
Class :character	1st Qu.:2021	Class :character	1st Qu.: 1.000	1st Qu.: 9.000
Mode :character	Median :2021	Mode :character	Median : 1.000	Median : 9.000
	Mean :2021		Mean : 1.551	Mean : 9.799
	3rd Qu.:2021		3rd Qu.: 2.000	3rd Qu.: 9.000
	Max. :2021		Max. :143.000	Max. :98.000

towing_and_articulation	vehicle_manoeuvre_historic	vehicle_manoeuvre
Min. :-1.00	Min. :-1.00	Min. :-1.00
1st Qu.: 0.00	1st Qu.: 8.00	1st Qu.: 8.00
Median : 0.00	Median :18.00	Median :19.00
Mean : 0.23	Mean :19.73	Mean :20.33
3rd Qu.: 0.00	3rd Qu.:18.00	3rd Qu.:19.00
Max. : 9.00	Max. :99.00	Max. :99.00

vehicle_direction_from	vehicle_direction_to	vehicle_location_restricted_lane_historic
Min. :-1.000	Min. :-1.000	Min. :-1.000
1st Qu.: 2.000	1st Qu.: 2.000	1st Qu.: 0.000
Median : 5.000	Median : 5.000	Median : 0.000
Mean : 4.389	Mean : 4.438	Mean : 7.936
3rd Qu.: 7.000	3rd Qu.: 7.000	3rd Qu.: 0.000
Max. : 9.000	Max. : 9.000	Max. :99.000

vehicle_location_restricted_lane	junction_location	skidding_and_overturning
Min. :-1.000	Min. :-1.000	Min. :-1.000
1st Qu.: 0.000	1st Qu.: 0.000	1st Qu.: 0.000
Median : 0.000	Median : 1.000	Median : 0.000
Mean : 7.933	Mean : 2.622	Mean : 0.928
3rd Qu.: 0.000	3rd Qu.: 6.000	3rd Qu.: 0.000
Max. :99.000	Max. : 9.000	Max. : 9.000

hit_object_in_carriageway	vehicle_leaving_carriageway	hit_object_off_carriageway
Min. :-1.000	Min. :-1.00	Min. :-1.0
1st Qu.: 0.000	1st Qu.: 0.00	1st Qu.: 0.0
Median : 0.000	Median : 0.00	Median : 0.0
Mean : 8.197	Mean : 1.01	Mean : 8.1
3rd Qu.: 0.000	3rd Qu.: 0.00	3rd Qu.: 0.0
Max. :99.000	Max. : 9.00	Max. :99.0

first_point_of_impact	vehicle_left_hand_drive	journey_purpose_of_driver_historic
-----------------------	-------------------------	------------------------------------

```

Min.      :-1.00      Min.      :-1.000      Min.      :-1.00
1st Qu.:  1.00      1st Qu.:  1.000      1st Qu.:  3.00
Median :  1.00      Median :  1.000      Median :  6.00
Mean     :  2.03      Mean     :  1.456      Mean     :  4.68
3rd Qu.:  3.00      3rd Qu.:  1.000      3rd Qu.:  6.00
Max.     :  9.00      Max.     :  9.000      Max.     :  6.00

journey_purpose_of_driver sex_of_driver      age_of_driver      age_band_of_driver
Min.      :-1.000      Min.      :-1.000      Min.      : -1.00      Min.      :-1.000
1st Qu.:  6.000      1st Qu.:  1.000      1st Qu.:  22.00      1st Qu.:  5.000
Median :  6.000      Median :  1.000      Median :  34.00      Median :  6.000
Mean     :  4.897      Mean     :  1.494      Mean     :  34.68      Mean     :  5.797
3rd Qu.:  6.000      3rd Qu.:  2.000      3rd Qu.:  50.00      3rd Qu.:  8.000
Max.     :  7.000      Max.     :  3.000      Max.     :101.00      Max.     :11.000

engine_capacity_cc propulsion_code      age_of_vehicle      generic_make_model
Min.      :  -1      Min.      :-1.000      Min.      :-1.000      Length:186443
1st Qu.:   49      1st Qu.:  1.000      1st Qu.:  0.000      Class :character
Median : 1388      Median :  1.000      Median :  5.000      Mode  :character
Mean     : 1358      Mean     :  1.085      Mean     :  6.001
3rd Qu.: 1950      3rd Qu.:  2.000      3rd Qu.:11.000
Max.     :29980      Max.     :12.000      Max.     :91.000

driver_imd_decile lsoa_of_driver      scooter_flag      driver_distance_banding
Min.      :-1.000      Length:186443      Min.      :0.000000      Min.      :-1
1st Qu.:  1.000      Class :character      1st Qu.:0.000000      1st Qu.: -1
Median :  4.000      Mode  :character      Median :0.000000      Median : -1
Mean     :  3.843      Mean     :0.007321      Mean     :0.007321      Mean     : -1
3rd Qu.:  7.000      3rd Qu.:0.000000      3rd Qu.:0.000000      3rd Qu.: -1
Max.     :10.000      Max.     :1.000000      Max.     :1.000000      Max.     : -1

```

Appendix 19 Head of vehicle dataset

```
> head(vehicle_dataset) #inspect the dataset
  collision_index collision_year collision_ref_no vehicle_reference vehicle_type
1    2021371073032         2021      371073032             1           9
2    2021010330300         2021      010330300             1           3
3    2021010331714         2021      010331714             1           3
4    2021471076364         2021      471076364             2           9
5    2021451012299         2021      451012299             2           9
6    2021131038600         2021      131038600             2           9
  towing_and_articulation vehicle_manoeuvre_historic vehicle_manoeuvre
1                      0                      18                      19
```

```

2                      0                      9                      9
3                      0                      99                     99
4                      0                      17                     19
5                      0                      4                      4
6                      0                      2                      2
  vehicle_direction_from vehicle_direction_to vehicle_location_restricted_lane_historic
1                      7                      3                      0
2                      3                      1                      0
3                      9                      9                      99
4                      7                      4                      0
5                      7                      3                      0
6                      0                      0                      0
  vehicle_location_restricted_lane junction_location skidding_and_overturning
1                      0                      0                      1
2                      0                      8                      5
3                      99                     9                      9
4                      0                      1                      0
5                      0                      0                      0
6                      0                      0                      0
  hit_object_in_carriageway vehicle_leaving_carriageway hit_object_off_carriageway
1                      0                      1                      0
2                      0                      0                      0
3                      99                     9                      99
4                      0                      0                      0
5                      0                      0                      0
6                      0                      0                      0
  first_point_of_impact vehicle_left_hand_drive journey_purpose_of_driver_historic
1                      4                      1                      5
2                      4                      1                      2
3                      1                      1                      2
4                      1                      1                      5
5                      2                      1                      5
6                      2                      1                      5
  journey_purpose_of_driver sex_of_driver age_of_driver age_band_of_driver
1                      6                      1          86          11
2                      2                      1          23           5
3                      2                      1          27           6
```

4	6	1	70	10	
5	6	2	42	7	
6	6	2	32	6	
	engine_capacity_cc	propulsion_code	age_of_vehicle	generic_make_model	driver_imd_decile
1	6200	1	48	-1	9
2	124	1	6	-1	9
3	124	1	7	-1	3
4	1968	2	14	DODGE CALIBER	3
5	1998	1	13	DODGE CALIBER	2
6	1798	1	13	DODGE CALIBER	8
	lsoa_of_driver	escooter_flag	driver_distance_banding		
1	E01021681	0	-1		
2	E01004079	0	-1		
3	E01000535	0	-1		
4	E01022904	0	-1		
5	E01016117	0	-1		
6	E01010861	0	-1		

Appendix 20 Head of accident vehicle metrics

car	motorcycle	bus
TRUE	FALSE	FALSE
FALSE	TRUE	FALSE
FALSE	TRUE	FALSE
TRUE	FALSE	FALSE
TRUE	FALSE	FALSE
TRUE	FALSE	FALSE

	hgv	bike	tractor	tram	mobility_scooter	other_vehicle	reversing	parking	waiting
1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
2	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
4	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
5	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
6	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
stopping moving_forward turning changing_lane overtaking unknown_manoeuvre no_junction									
1	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE
2	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE
3	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	TRUE
4	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
5	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE
6	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE
approaching cleared leaving entering mid_junction no_skidding skidded overturned									
1	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
2	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE
3	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
4	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
5	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
6	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE
jackknifed did_not_hit_object hit_object young_driver middle_aged_driver old_driver									
1	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE
2	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE
3	FALSE	TRUE	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE
4	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	TRUE
5	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE	FALSE
6	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	FALSE
old_car young_car petrol oil electric steam gas diesel other_propulsion									
1	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
2	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
3	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
4	FALSE	TRUE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE
5	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
6	FALSE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE